SINGLE CELL RNA – SEQ ANALYSIS

Analysis of a human tissues from PanglaoDB
BONE MARROW

TRANSCTIPTOMIC PROJECT
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BONE MARROW

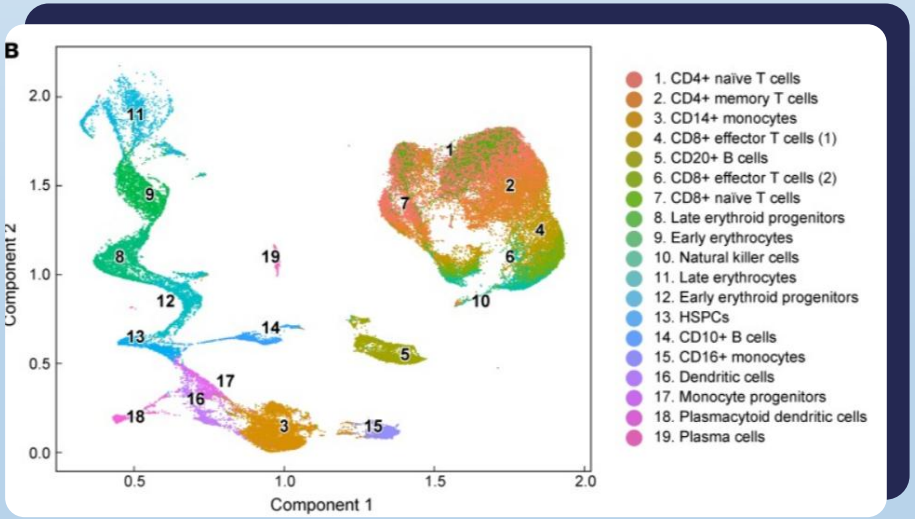
PanglaoDB

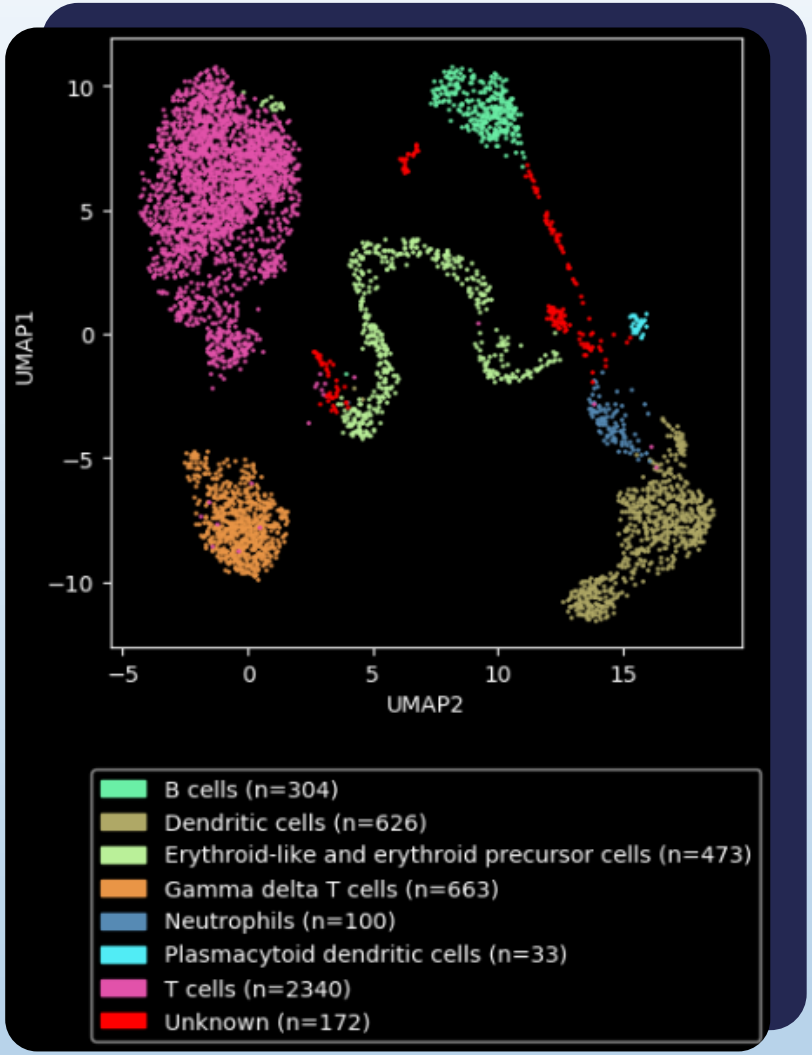
Main Features:

- Species: Homo Sapiens
- Number of cells: 5273
- Tissue: Bone Marrow
- Primary adult tissue
- Protocol: 10x Chromium

Publication:

Human bone marrow assessment by single-cell RNA sequencing, mass cytometry, and flow cytometry



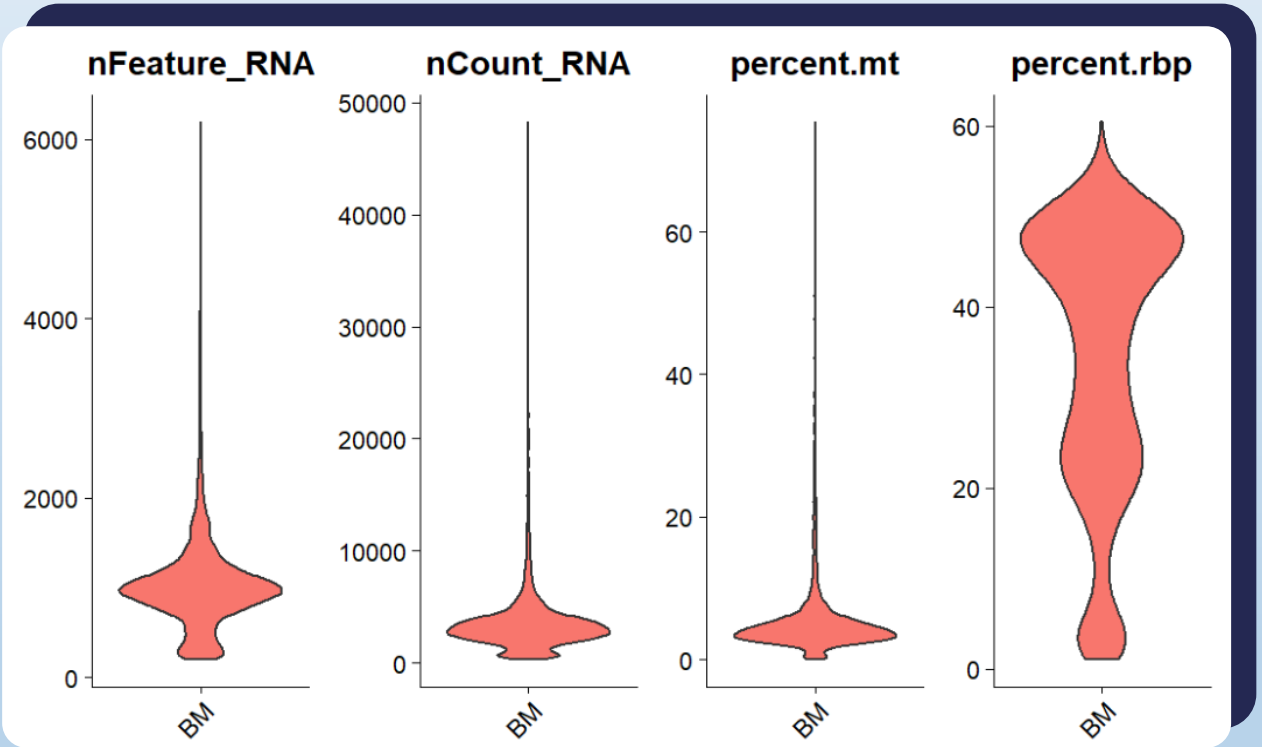


panglaodb

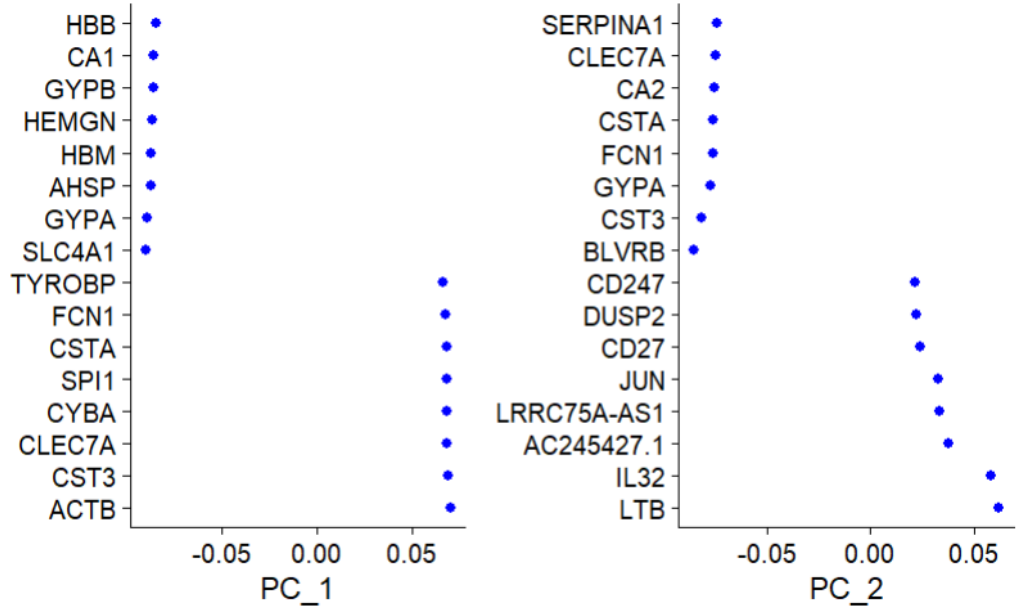
CELL QUALITY CONTROL

- **Quality control thresholds applied to retain cells:**
  - Min Gene Count: nFeature_RNA > 200
  - Max Gene Count: nFeature_RNA < 2500
  - Mitochondrial Content: percent_mt < 5%

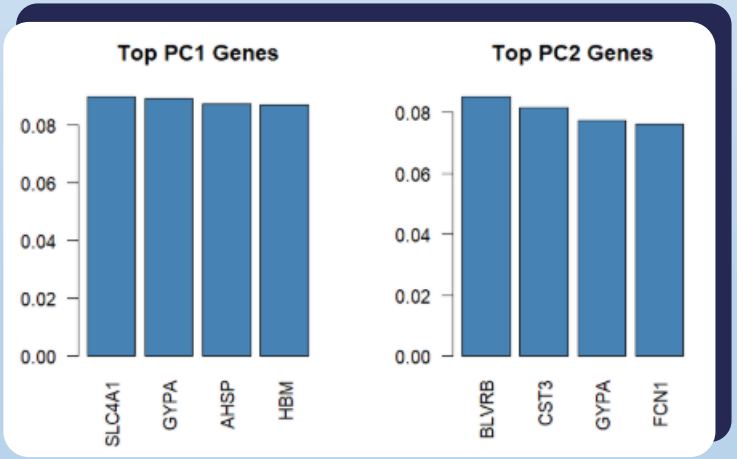
- Cells discarded: 1804
- Cells remaining after QC: 3433



PRINCIPAL COMPONENT ANALYSIS

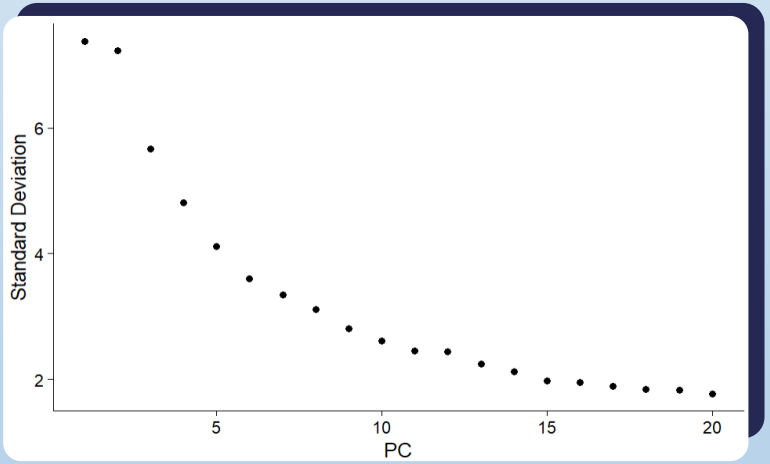


The first two Principal Component clearly separates biological lineages.



- The Elbow Plot shows an inflection point around **10-12 PCs**.
  - 10 PCs, Resolution 0.5 → Resulted in **15 Clusters**
- The cumulative variance to capture 75% of the signal, resulting in **17 PCs**.
  - 17 PCs, Resolution 0.5 → Resulted in **14 Clusters**

  This was the chosen threshold to avoid losing biological information.



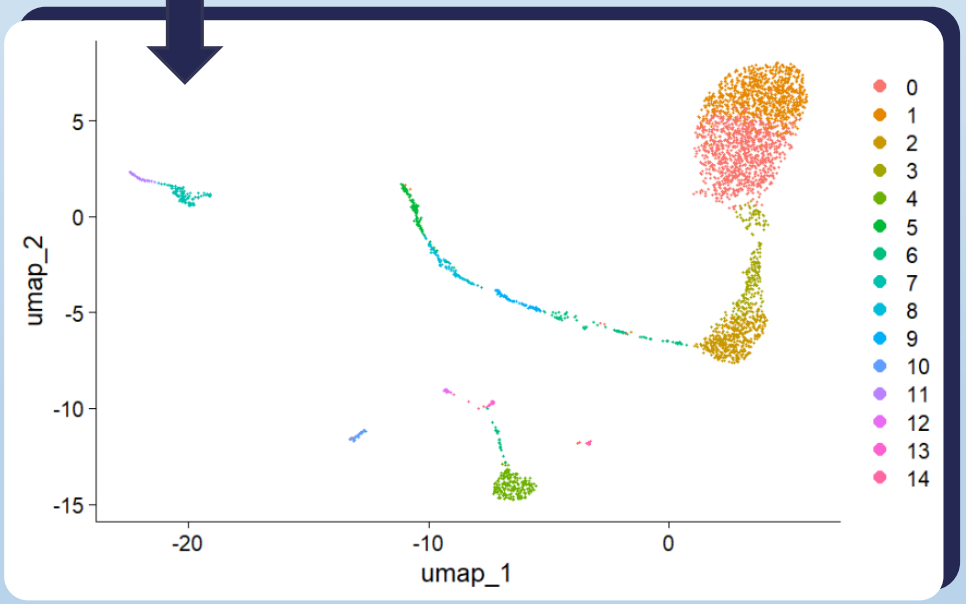
CLUSTERING

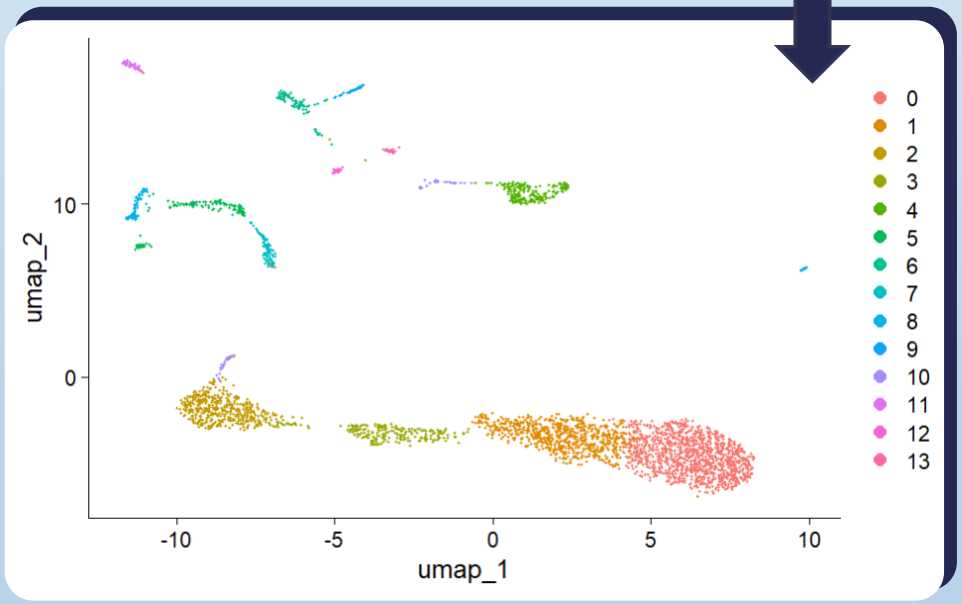
17 Principal Components → 14 clusters

0	1	2	3	4	5	6	7	8	9	10	11	12	13
1168	725	444	252	214	129	101	98	71	64	61	45	31	30

10 Principal Components → 15 clusters

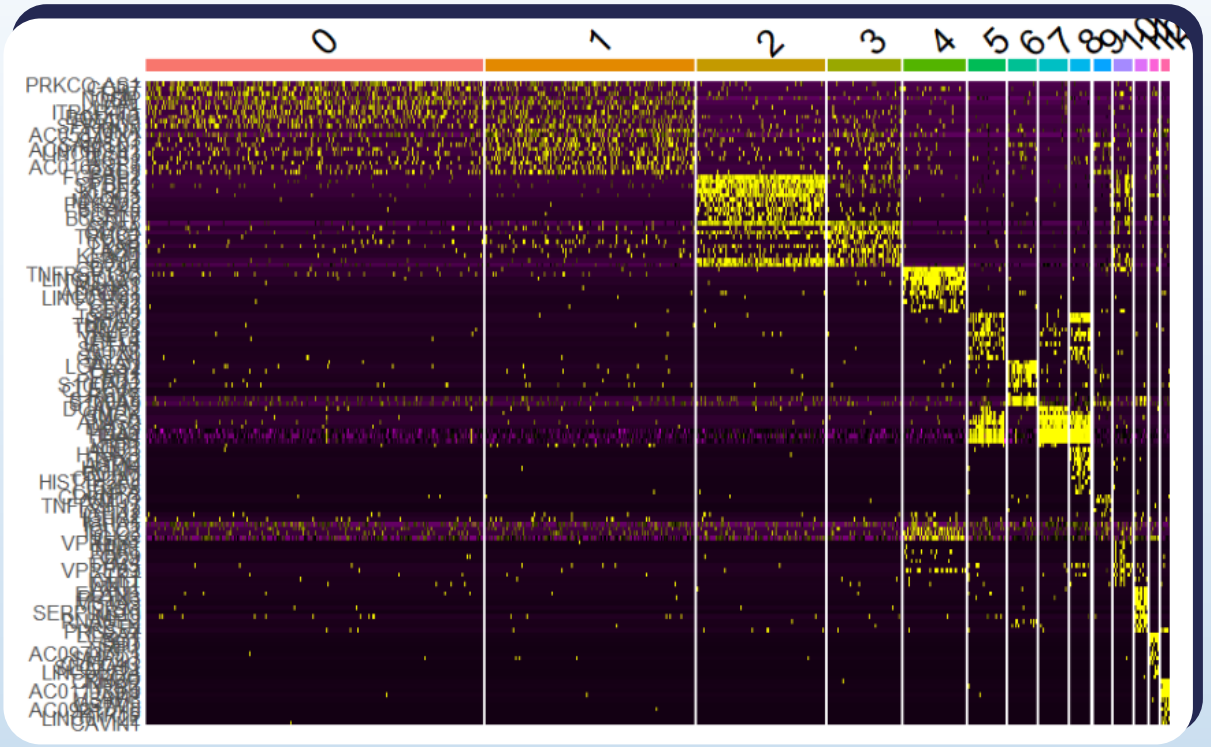
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1034	875	441	231	213	105	101	100	91	71	46	43	30	29	23

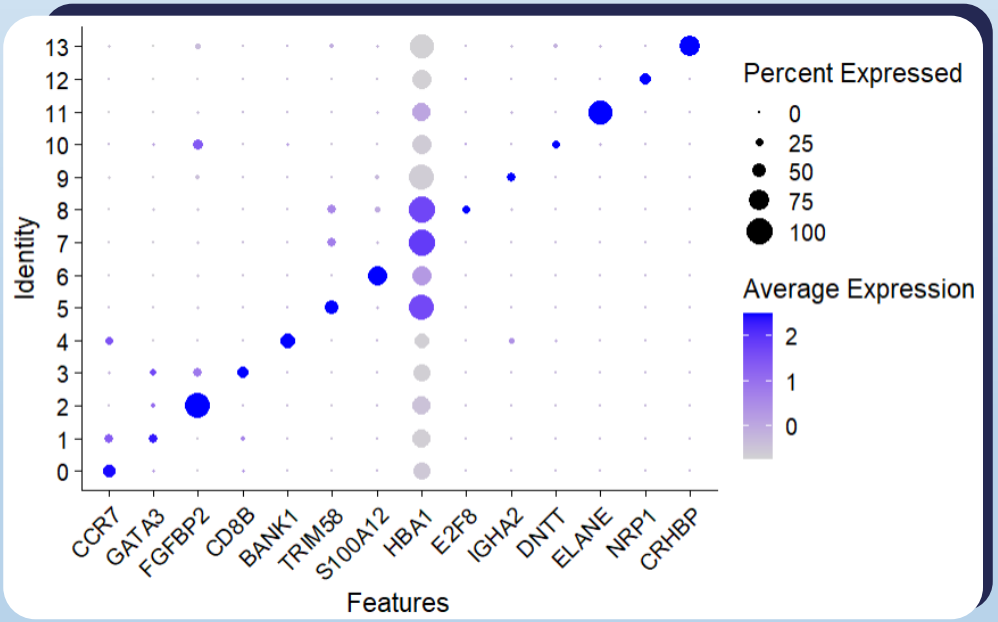




MARKER GENES

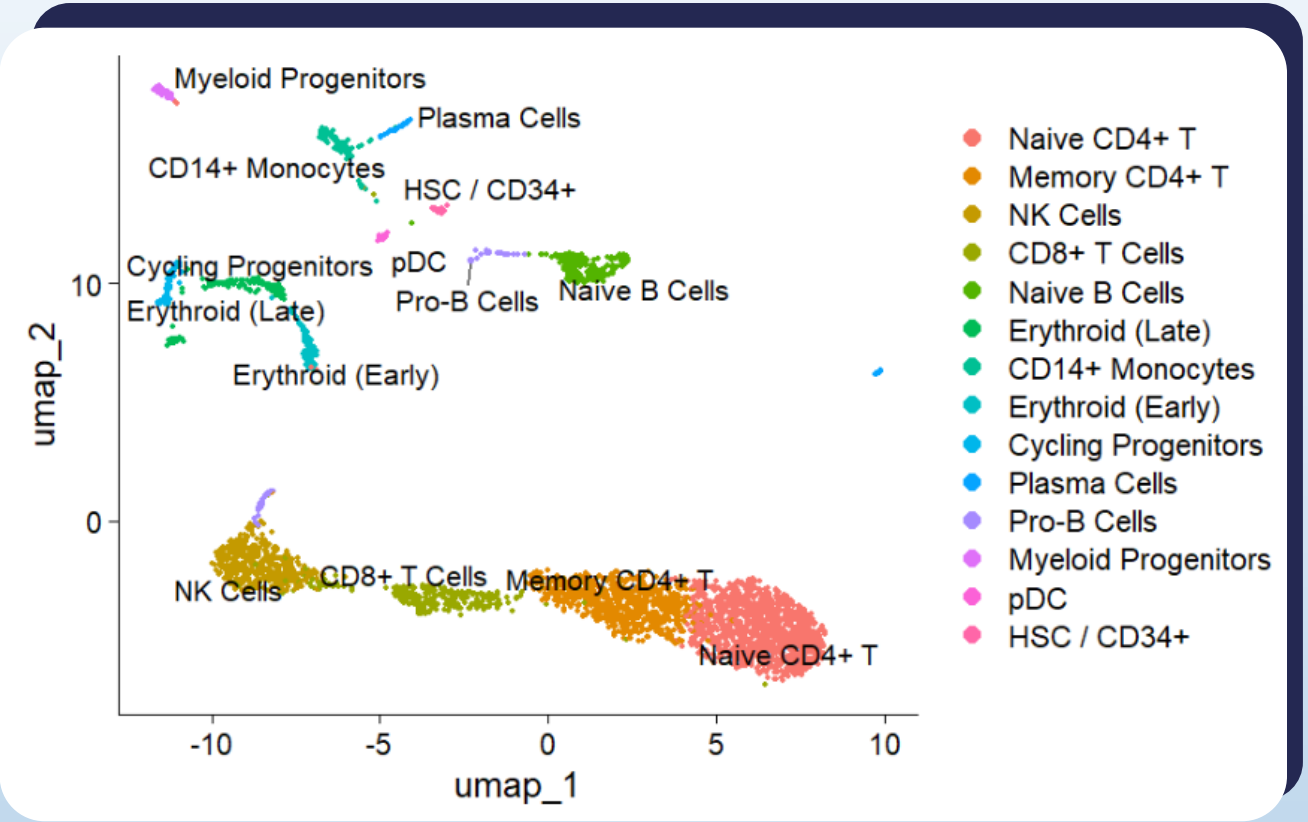
The heatmap shows that some clusters seem to share part of their over-expressed genes.
Clusters 0 and 1 show a broader, overlapping band, as both are composed of T cells at different stages of maturation.
Similarly, **Clusters 5, 7, and 8** represent erythroid cells at distinct developmental stages.





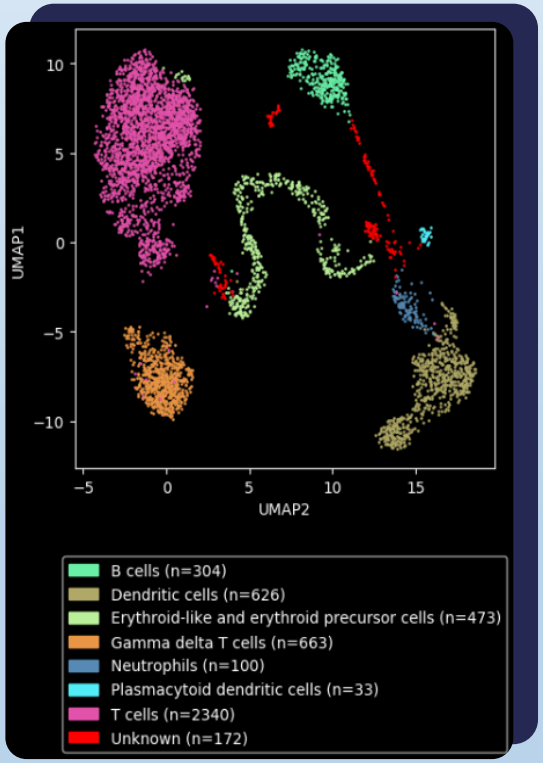
This dotplot illustrates a single representative marker gene for each cluster, curated from the top differentially expressed (DE) genes to optimally support cell type annotation.

CONCLUSIONS



Differences from PanglaoDB

- Manual annotation proved significantly more precise and biologically accurate.
- PanglaoDB oversimplified merging 2,340 cells into a generic "T cells" group.
- The automated database completely missed critical populations like Plasma cells and Hematopoietic Stem Cells.



Differences from the publication

- The analysis successfully identified all the major bone marrow cell lineages reported in the original paper.
- The publication found 19 clusters compared to my 14, achieving a higher level of granularity through sub-clustering.